



TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2015 Answers

BIOLOGY
Higher 2
Paper 2 Core Paper

9648/02
Friday, 28 August
2 hours

QUESTION 1

- (a) (i) Identify organelles **A** and **B**. [2]

A: Rough Endoplasmic Reticulum

B: Golgi apparatus

- (ii) Describe how a biomolecule synthesized in **A** is transported to **B**. [2]

Proteins synthesized at ribosomes pass through the membrane of the rough Endoplasmic Reticulum (ER) into the ER lumen. ER vesicle containing the proteins buds off, travels along microtubules of the cytoskeleton to cis-face of Golgi apparatus, fuses with the Golgi membrane.

- (b) The passage of most molecules through cell membrane is regulated by proteins. State the roles of these proteins. [2]

Channel protein provides a hydrophilic channel for hydrophilic molecules to pass through the cell membrane.

Carrier protein transports molecules across the membrane via a conformational change/ allows molecules with shape complementary to its binding site/ substances that can bind to it to enter.

- (c) Numbers **1** to **2** in **Fig. 1.2** represent a sequence of events occurring in the cell. Describe the process that is taking place in **Fig. 1.2**. [4]

Vesicle travel along microtubules of the cytoskeleton towards the cell membrane. Vesicle fuses with the cell membrane. Contents of the vesicle is released out of the cell. Exocytosis occurs. Membrane of vesicles contributes to the cell membrane/ forms the cell membrane, resealing the gap in the plasma membrane.

[Total: 10]

QUESTION 2

- (a) Explain how the stability of the DNA structure is maintained. [3]

2 polynucleotide strands are anti-parallel which facilitate the formation of hydrogen bonds between complementary nitrogenous bases that project sideways from the sugars.

Adenine always base pair with Thymine and Cytosine always base pair with Guanine and vice versa; a pyrimidine on one strand always base-pairs with a purine on the other strand.

Width between the 2 sugar-phosphate backbones is constant. There is hydrophobic interactions between stacked bases.

Hydrophobic bases are found at the interior of the helix while hydrophilic sugar-phosphate backbone is found at the exterior of the helix.

- (b) (i) State the name of the enzyme that is found in ribosomes and describe the reaction it catalyses. [2]

Peptidyl transferase catalyzes the formation of peptide bonds between amino acids via condensation reaction.

- (ii) Suggest how the RNA molecules form the catalytic site of the ribosome. [2]

RNA polynucleotide strands / molecules fold into compact 3D shape forming an active site with a specific shape that is complementary to the shape of the amino acids/ the growing peptide chain and the aminoacyl-tRNA complex. The shape is maintained by hydrogen bonds between ribonucleotides and interactions with ribosomal proteins.

Lactimidomycin, a potential anti-cancer drug, is found to selectively inhibit translation. Laboratory studies involving lactimidomycin have resulted in an accumulation of dipeptides in the cells.

- (c) Suggest how lactimidomycin works to inhibit translation. [1]

It prevents the translocation of ribosomes along the mRNA by binding to the empty "E" site/ mRNA before the start codon.

[Total: 8]

QUESTION 3

- (a) Explain the difference between the appearance of euchromatin and heterochromatin. [2]

Heterochromatin appears darker/denser than euchromatin because heterochromatin is more condensed/ more tightly coiled/ more compact than euchromatin.

- (b) (i) State two forms of epigenetic modifications that can prevent gene expression in eukaryotes. [2]

1. DNA methylation
2. Histone deacetylation

- (ii) Outline how the two forms of epigenetic modifications stated in (a)(i) can prevent transcription. [4]

Methyl groups (-CH₃) added to lysine residues found in the histone tails or cytosine bases in DNA. Proteins that bind to methylated DNA recruit histone deacetylation enzymes.

Acetyl groups (-COCH₃) are removed from the acetylated lysine residues found in the histone tails. The affinity of histone tails and the affinity of the entire histone complex for DNA are restored. This tightened the winding of DNA around nucleosome core particle and further packing of the chromatin fiber. Transcription factors cannot bind to the promoter/ transcription initiation complex cannot assemble at the promoter/ RNA pol cannot bind to promoter.

- (c) Explain the role of transcription factors in the control of gene expression. [3]

Binding of transcription factors to proximal control elements and activators to enhancers help to increase the rate of transcription by stabilizing the transcription initiation complex, which helps RNA polymerase to achieve promoter clearance. Binding of transcription factors to the promoter enables the formation of the transcription initiation complex/ attachment of RNA polymerase to promoter.

- (d) Suggest and explain the likely observation which will be made on the occupancy of transcription factors on certain genes in cancerous cells. [2]

High occupancy of transcription factors increased rate of transcription of oncogenes to increase the rate of cell division.

[Total: 13]

QUESTION 4

- (a) Describe how the structure of telomeric DNA helps telomeres carry out its function.

[3]

Function of telomere

ANY ONE:

Prevent the ends of chromosomes from being recognized as broken chromosomes, by the cellular repair machinery, which could trigger apoptosis (programmed cell death)

OR

Protect the chromosomal ends from degradation and from joining to other chromosomes

OR

Prevent the loss of genes near the ends of the DNA molecule

The 3' end at each telomere is always slightly longer than the 5' end/ protruding single-stranded 3' end/ 3' overhang. 3' protruding end has been shown to loop back

to tuck its single-stranded terminus into the duplex DNA/ forming t-loop aided by specialized proteins

- (b) State and explain the significance of an increase in the rate of telomere shortening. [2]

It decreases the lifespan of cells

OR

It reduces the number of times a cell can divide.

Cumulative loss of telomeric DNA occurs faster which signals cell division to cease earlier in a normal cell.

OR

Telomere is shortened to its critical length faster therefore apoptosis is triggered earlier.

- (c) Outline how genomic instability and elevated telomerase activity can contribute to the development of cancer. [4]

Mutation/ gain of function mutation in one proto-oncogene and loss of function mutation in several tumour suppressor genes increases the chances of forming a benign tumour/ uncontrolled cell growth and division which develops into a malignant tumour with further mutations (that contribute towards angiogenesis and metastasis)

Telomere is not shortened to its critical length. Cells are able to divide for a longer period of time/ do NOT undergo apoptosis. This makes it more probable for cells to accumulate mutations across generations of cells.

[Total: 9]

QUESTION 5

- (a) Identify structures A and B.

[2]

Structure A: Capsid/ nucleocapsid/ head (of bacteriophage)Structure B: Non-contractile tail (of bacteriophage)

- (b) Using a named example, describe the processes that lead to the formation of prophage. [3]

The tail fibre of the lambda phage binds to specific protein receptor sites on the bacterial cell wall/ membrane/ surface. Upon binding, the phage injects its DNA into the cytoplasm of the bacterium cell. Within the host cell, the linear phage DNA circularises. The phage undergoes the lysogenic cycle whereby the circular phage DNA integrates into a specific site on the host bacterial chromosome, thus forming a prophage.

- (c) (i) Describe the effect of lactose on the
- lac*
- operon.

[3]

In the presence of lactose, lactose is converted to allolactose by β -galactosidase. Allolactose binds to the *lac* repressor at an allosteric site and changes shape of *lac* repressor to an inactive form. There is removal of negative control. As the shape of the *lac* repressor is NO longer complementary to the shape of the operator, the inactive repressor does NOT bind to the OPERATOR, thus allowing RNA polymerase to bind to PROMOTER. The genes of the operon are transcribed and translated. β -galactosidase, *lac* permease and β -galactoside transacetylase are produced.

- (ii) Explain why the number of bacteriophages in the medium increased after the addition of lactose. [2]

As the presence of lactose switches on the *lac* operon, the prophage genome that lies exactly in between *lacZ* and *lacY* genes of the bacterial *lac* operon was also transcribed, forming new phage proteins. When the lytic cycle is initiated, the proteins self-assemble to form phage heads, tails and tail fibres. New bacteriophages are formed when the nucleocapsids, tail and tail fibres assemble together. When the bacterial cell lyses, new bacteriophages are released into the medium, thus causing the increase in the number of bacteriophages in the medium.

[Total: 10]

QUESTION 6

(a) State the meaning of the term *epistasis* in this context.

[3]

The pigmented phenotype is the result of the interaction between 2 different genes. Pigments are produced in the presence of at least one dominant allele for each gene. The genes each code for a respective enzyme in a biochemical pathway that leads to the production of pigment. When EITHER gene is homozygous recessive, it masks the phenotypic expression of the OTHER gene.

(b) Use the symbols, **A**, **a** and **B**, **b** to draw a genetic diagram to explain the results shown in the second generation.

A represents the allele that codes for the first enzyme in the pathway for pigment production

a represents the allele that codes for a non-functional enzyme

B represents the allele that codes for the second enzyme in the pathway for pigment production

b represents the allele that codes for a non-functional enzyme

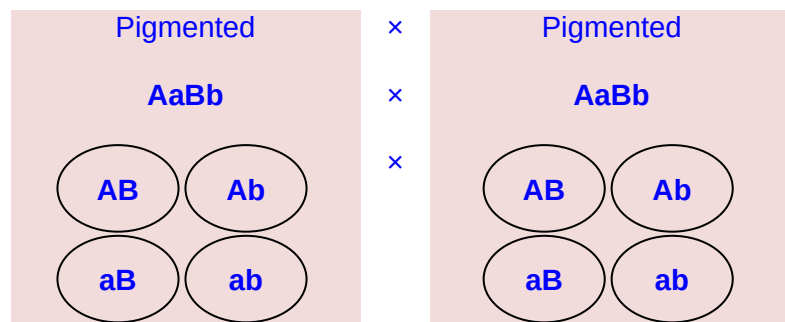
The allele that codes for an enzyme (**A**) is dominant to the allele that codes for a non-functional enzyme (**a**).

The allele for enzyme (**B**) is dominant to the allele that codes for a non-functional enzyme (**b**).

Offspring 1 phenotype:

Offspring 1 genotype:

Offspring 1 gametes:



Punnett Square to show the fusion of gametes:

♀ \ ♂	AB	Ab	aB	ab
AB	AABB Pigmented	AABb Pigmented	AaBB Pigmented	AaBb Pigmented
Ab	AABb Pigmented	AAbb Albino	AaBb Pigmented	Aabb Albino
aB	AaBB Pigmented	AaBb Pigmented	aaBB Albino	aaBb Albino
ab	AaBb Pigmented	Aabb Albino	aaBb Albino	aabb Albino

Offspring 2 genotype:

AABB
AABb
AaBB
AbBb

:

AAbb
Aabb
aaBB
aaBb
aabb

Offspring 2 phenotype:

Pigmented

:

Albino

Phenotypic ratio of Offspring 2:

9

:

7

Observed number:

740

:

561

[4]

A chi-squared test was carried out on the results of the second cross.

Phenotype	Observed number (O)	Expected number (E)	$\frac{(O-E)^2}{E}$
Pigmented	740	732	0.087
Albino	561	569	0.112
χ^2 value = 0.20			

Part of the critical values of the chi-squared distribution is shown below.

degrees of freedom	probability						
	0.995	0.975	0.9	0.5	0.1	0.05	0.01
1	.000	.000	.016	0.455	2.706	3.841	6.635

- (c) Explain how the chi-squared calculated value of 0.20 supports the statement that epistasis is the correct explanation of these results. [3]

For 1 degree of freedom, χ^2 value of 0.20 is less than 3.84 Probability that the deviation is due to chance is more than 0.05. The deviation is therefore statistically insignificant and due to chance. Epistasis is the correct explanation of these results.

[Total: 10]

[Turn over

QUESTION 7

- (a) With reference to Table 7.1, using the evidence from the non-coding regions, explain why *D. miranda* may be more closely related to *D. persimilis* than to *D. pseudoobscura*. [3]

The percentage of divergence of base sequence of *D. persimilis* from *D. miranda* is less compared to divergence of *D. pseudoobscura* from *D. miranda* for both non-coding regions. For region 1, the difference between *D. miranda* and *D. persimilis* is 2.4% while the difference from *D. pseudoobscura* is 2.8%. For region 2, the difference between *D. miranda* and *D. persimilis* is 7.3% while the difference from *D. pseudoobscura* is 8.1%. This means that *D. miranda* and *D. persimilis* share a more recent common ancestor / diverge from a common ancestor more recently. As a result, there was less time to accumulate mutations/ fewer mutations observed in the DNA sequences.

- (b) Explain why non-coding DNA is suitable for constructing phylogenetic relationships. [3]

Non-coding DNA are not expressed in phenotype/ do NOT code for protein product. Changes in non-coding DNA is neutral/ mutation is silent. Hence they do NOT confer selective advantage or disadvantage/ do NOT affect the fitness of the organism/ do NOT affect ability of organism to survive and produce viable offspring and are NOT subjected to natural selection. Mutations in non-coding DNA can be accumulated/ can be passed on and compared. Number of mutations can indicate how long 2 species separated/ how long ago they became separate species.

- (c) State the importance of variation in the coding region for evolution to occur. [2]

It may result in formation of new alleles which codes for new phenotype / gene product (May confer selective advantage to the organism OR May allow better adaptation to the environment) in the event of sudden/ drastic change. It increases variation in the population allowing natural selection to operate.

- (d) Outline how the species *D. pseudoobscura* could have evolved from a population of *D. miranda*. [4]

Allopatric speciation took place. Mountain leads to presence geographical isolation/ physical, barrier. Reproductive isolation/ NO interbreeding/ NO gene flow between populations. Different selection pressure present. Only selective advantage individuals will survive & live to reproductive age to produce viable offspring in that location. Increase in individuals with favourable alleles for survival. Gene pools of isolated populations change in genetic composition/ gene pools become distinct OR accumulated so much modifications. NO longer able to interbreed to form fertile and viable offspring.

[Total: 12]

QUESTION 8

(a) With reference to Table 8.1,

(i) state the difference between the blood glucose concentrations of the diabetic and non-diabetic individuals between 3 pm and 5 pm; [1]

Between 3 and 5pm, the concentration of blood glucose in the diabetic decreased from 219mg/100ml of blood to 105mg/100ml of blood, whereas the concentration of blood glucose in the non-diabetic remained relatively constant between 90 and 91mg/100ml of blood.

(ii) explain why there is a difference in blood glucose concentrations of the diabetic and non-diabetic individuals between 3 pm and 5 pm. [2]

In the diabetic person, there is NO insulin produced. Cellular responses are NOT triggered in target cells such as muscle and liver cells to increase glucose uptake OR to convert excess glucose to glycogen. Subsequent decrease in blood glucose is due to uptake of glucose by muscle cells during exercise from the blood.

In contrast, glucagon is produced/ secreted in non-diabetic individual which helps to maintain blood glucose level at the set point of about 90mg/100ml of blood when blood glucose level drops below set point.

OR

In non-diabetic individuals, presence of stored glycogen in muscle cell enables the individual to convert glycogen to glucose for use in cellular respiration. Therefore, blood glucose is NOT used and blood glucose concentration stays relatively constant at set point.

(b) Describe how phosphorylated IRS-1 decreases blood glucose concentration in a non-diabetic. [3]

Phosphorylated IRS-1 protein then phosphorylate and activate PI3-kinase. PI3-kinase then phosphorylates PIP₂ to produce PIP₃, which then phosphorylates and activates Akt protein. This forms a phosphorylation cascade in the pathway, thus resulting in a cellular response to the signal. This eventually leads to the translocation of vesicles with glucose transporters (GLUT4) to the plasma membrane, INCREASING the amount of glucose transporters (GLUT4) on the plasma membrane. This increases the permeability of cell membrane to glucose, thus increasing uptake of glucose from the blood by cells and therefore, blood glucose concentration decreases.

(c) Describe **one** significance of having a cell signaling system. [2]

It is an efficient cascade system whereby a very low concentration of signal molecules, insulin, that bind to plasma membrane receptors initially is sufficient can result in a much greater cellular response through multiple signal cascades (Signal amplification).

OR

The binding of the ligand/ signal to receptors on the cell membrane is able to activate genes in the nucleus (within the cell). Therefore, the ligand does not need to cross the phospholipid bilayer to effect an cellular response.

[Total: 8]

SECTION B

- 9 (a) Outline the molecular structure of phospholipids in relation to their function in cell membranes. [6]

Function of phospholipids	Structure	How structure allows it to perform function
The phospholipid bilayer functions as a hydrophobic boundary between the aqueous interior and exterior of the cell.	Phospholipids consists of <u>a glycerol attached to a phosphate head and two hydrocarbon tails from two fatty acids.</u> The phosphate head carries an <u>electrical charge</u> and is <u>hydrophilic.</u> The <u>hydrocarbon tails are neither charged nor polar</u> and hence <u>hydrophobic.</u> Phospholipids are <u>amphipathic.</u>	In an aqueous environment, <u>phospholipid molecules assemble themselves into a bilayer.</u> The <u>hydrophilic phosphate heads faces outwards</u> and make <u>contact with the aqueous environment on either sides.</u> The <u>hydrophobic tails face inwards</u> and are <u>sandwiched / buried between the hydrophilic heads.</u> The <u>hydrophilic region allows the hydrophobic boundary to exist in an aqueous environment.</u>

- (b) Discuss the roles of NAD and NADP. [6]

Both NAD and NADP are electron carriers that are used in respiration and photosynthesis respectively. NAD is reduced during glycolysis, link reaction and Krebs cycle to form a total of 10 NADH. NADP is reduced during photophosphorylation to form NADPH. NADH donates electrons to the electron transport chain during oxidative phosphorylation. During oxidative phosphorylation, NADH to be oxidized to form NAD. NADP is the final electron acceptor of the electron transport chain in non-cyclic photophosphorylation. Both are needed to drive ATP synthesis via chemiosmosis. In particular, oxidation of 10 NADH drives the synthesis of 30 ATP. NADPH is oxidized during Calvin cycle for glucose synthesis.

[Turn over

(c) Compare oxidative phosphorylation and photophosphorylation.

[8]

Similarities:

In both oxidative phosphorylation and photophosphorylation.

(S1) ATP is synthesized.

(S2) the flow of electrons down the ETC is used to create a proton gradient across a membrane.

(S3) there is a flow of electrons down electron carriers of decreasing energy levels.

(S4) protons diffuse down its gradient via the proton channel in the ATP synthase.

(S5) energy released from the diffusion of protons is used to drive ATP synthesis.

[Any other valid points; max 4 marks]

Differences:

Factors	oxidative phosphorylation	Photophosphorylation
(D1) Electron carriers involved	FADH, NADH	NADP
(D2) Function of electron carriers	Electron donors	Final electron acceptor
(D3) Source of electrons/ electron donors	FADH, NADH	H ₂ O
(D4) Final electron acceptor	Oxygen	NADP
(D5) Products	FAD, NAD, ATP and H ₂ O	NADPH, ATP and oxygen
(D6) Involvement of light	Light is not needed.	Light is needed for photolysis of water.

[Turn over

- 10 (a)** Explain how the structure of collagen and haemoglobin are related to their function. [8]

State function of collagen:

To provide strength and resilience to tissues, and to help tissues stretch.

Level of protein structure	Structural features of collagen	Characteristics that it confers on collagen so that it can carry out its function
Primary Structure	Each polypeptide chain has about 1000 amino acid residues, and consists <u>mainly of a REPEAT of glycine, proline and hydroxyproline residues.</u>	Glycine's <u>small R group</u> fits into the centre of the triple helix and <u>allows close association of the three polypeptide chains.</u> OR The <u>rigid peptide bond formed with proline</u> residues helps to <u>stabilize the triple helix.</u>
Secondary Structure	<u>The localised, repetitive folding of 1 polypeptide chain</u> enables the <u>hydrophobic R-groups</u> of proline and hydroxyproline residues are on the <u>exterior</u> of the triple helix.	This renders collagen <u>insoluble</u> , making it a suitable structural protein.
Quaternary Structure	<u>Three helical chains wound round one another to form a triple helix structure</u> with <u>interchain hydrogen bonds</u> formed between the <u>-NH groups on the glycine residues on one chain</u> , and the <u>-CO groups of the proline of an adjacent chain.</u>	Extensive interchain hydrogen bonds <u>stabilize the fibrous structure of collagen</u> and cause it to be <u>rigid</u> . This enables collagen to have <u>high tensile strength.</u>
	The tropocollagen molecules lie in <u>staggered array in parallel bundles</u> , with <u>covalent cross-links between C and N terminals.</u> This forms the <u>collagen fibrils</u> , which further assemble to form <u>collagen fibers</u>	This gives <u>additional rigidity and tensile strength</u> to collagen.

(HF) It functions in the transport of oxygen.

Level of protein structure	Structural features of haemoglobin	Characteristics that it confers on haemoglobin so that it can carry out its function
Tertiary Structure	In each subunit, the <u>hydrophobic amino acid residues are buried in the interior</u> of the folded structure while the bulk of the <u>hydrophilic amino acid residues are on the exterior.</u>	This makes globular haemoglobin molecule <u>soluble in water</u> , which makes it suitable as a transport protein.

	Each subunit contains a haem group, which consists of a <u>porphyrin ring and an iron ion (Fe^{2+})</u> .	Fe^{2+} ion is able to <u>bind reversibly to oxygen</u> , accounting for the oxygen-carrying ability of haemoglobin.
Quaternary Structure	Each haemoglobin has 4 subunits, therefore it has <u>4 haem groups</u> .	<u>One haemoglobin molecule can carry up to 4 oxygen molecules</u> .
	The four subunits are held by various <u>non-covalent interactions</u> .	This allows for cooperative oxygen binding, where <u>binding of oxygen to the haem group of one globin subunit to trigger a conformation change in the other three globin subunits</u> . This <u>increases the affinity of the other globin subunits to O_2 and faster uptake of O_2 by subsequent haem groups in the same haemoglobin molecule</u> .

(b) Outline the main features of anaerobic respiration.

[6]

In respiration under anaerobic conditions, fermentation occurs to regenerate NAD^+ so that glycolysis can occur. NADH produced from glycolysis is oxidised by pyruvate to give either ethanol (alcohol fermentation) or lactate (lactate fermentation). Like glycolysis, both alcohol and lactate fermentation occur in the cytosol. Alcohol fermentation occurs in plants and yeast cells. Pyruvate is first decarboxylated to ethanal. The enzyme is pyruvate decarboxylase. Ethanal is then reduced by NADH to form ethanol. NAD^+ is regenerated. The enzyme involved is alcohol dehydrogenase. The ethanol remains unchanged even when aerobic conditions return. Lactate fermentation occurs in muscle cells of animals. Pyruvate is reduced by NADH to lactate. NAD^+ is regenerated. The enzyme involved is lactate dehydrogenase. When aerobic conditions return, lactate is transported to the liver and converted to pyruvate. Only a small amount of NAD^+ is regenerated from fermentation. This is sufficient for glycolysis to occur, but not for the subsequent processes. 2 NET ATP is produced during anaerobic respiration (glycolysis).

(c) Compare competitive and non-competitive inhibitors. [5]

Similarities:

S1 Both lower rate of enzyme activity [1]

S2 Both inhibitors can bind reversibly to the enzyme's active site. [1]

Differences

	COMPETITIVE INHIBITION	NON-COMPETITIVE INHIBITION
(D1) Site of binding of inhibitor	At the <u>active site</u> of the enzyme	At a region <u>other than the active site</u> of the enzyme
(D2) Structure of inhibitor	<u>Structurally similar</u> to the <u>substrate</u>	<u>NO structural resemblance</u> with the substrate
(D3) Effect of increasing substrate concentration on inhibition reaction rate	At high substrate concentrations, the <u>effects of inhibitor</u> are <u>NOT observed</u> .	At high substrate concentrations, the <u>inhibitory effects are still observed</u> .
(D4) Maximum rate of reaction	Can be <u>attained</u> if <u>substrate concentrations</u> are <u>sufficiently high</u> .	Can <u>never be attained</u> regardless of how high the substrate concentration is

✂ End of Paper ✂